

touch displays and found them to be too expensive. However, during their search they happened to find open source resources that provided a wealth of knowledge, designs and software to get started with building their own touch table. Starting off with the open source plans, they built a couple of touch table prototypes to try out their in-house software, mostly as a design experiment to explore the space within the context of the firm's software needs. When it worked well, they started the Playsurface project to share their hardware and software with a larger community as well as to provide a platform to enable further development.

## Electric cars for a greener world

<http://www.indiegogo.com/ZWheelz>

The EZ-EV Open Source Electric Kit Car is a very



easy-to-assemble three-wheel all-electric kit car that anybody can build and maintain. Gary Krysztopik, an electrical engineer, started working on electric vehicles inspired by the movie *Who Killed the Electric Car*. He has been working on electric cars full-time for about five years now. He is currently updating his prototypes and uploading them as open source. His prototypes are simple vehicles for basic transportation but can be upgraded, modified and maintained over the years as the design continues to evolve. The idea is that anybody can build one or buy parts and kits from anyone who wants to sell them. Krysztopik's idea of open sourcing a car design has many benefits: first, it would support local manufacturing; next, people get a car that they will always get parts for; plus, they can always change it to suit their needs due to the very modular design and the continuously emerging ideas from the community.

**The open twist:** While working on the electric kit car, the maker found it helpful that many of today's computer numerically controlled (CNC) machines and 3D printers are based on open source designs, making it possible to email files and build products locally. He also found it possible to do inexpensive rapid

prototyping with scale models using the same CAD model. So, for the car frame, he plans to cut fiberglass/honeycomb sheets with a CNC machine and stack them together, forming interlocking 3D shapes. This will be strong, light, easy to manufacture and assemble, besides being non-conductive and non-corrosive. All the CAD files, parts, materials, tools and machines are available online and he will continue to publish everything about the car at the ZWheelz Indiegogo campaign page. He also aims to create accurate, functional scale model kits for science and technology education in schools.

## A low-cost, do-it-yourself 3D printer

<http://www.eventorobot.com/>

Eventorobot is an open source 3D printer. The machine melts plastic filaments (ABS/ PLA) and builds objects, one layer at a time, from the bottom up. It features an easy-to-assemble, simple design, using a sturdy frame and fewer plastic parts compared to other such projects.

By using a single solid steel frame, it eliminates 40 per cent of the parts and costs that other 3D printers incur for the housing structure.

**The open twist:** The project is 100 per cent open source, and all the plans are available online. The blueprint, a complete hardware list with model numbers and links to the official websites, etc, are all available. Another interesting aspect is that 80 per cent of the parts can be self-replicated using the printer; so once you build a machine you can easily use the designs to print out



the replicable parts for your family or friends to use! By making the project open source, the founder expects that there will be several interesting attachments and upgrades from the community, in the future. 

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